User Journey: from raw data to an approved allocation plan

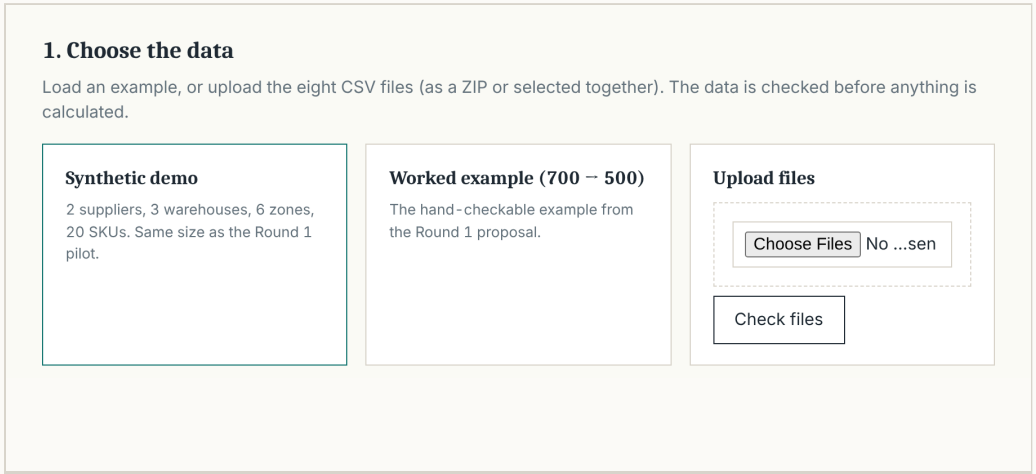
One complete end-to-end path through the running MVP, using the synthetic demo scenario.

ACTOR **Planner**, simulated single local user, accountable for the weekly plan.

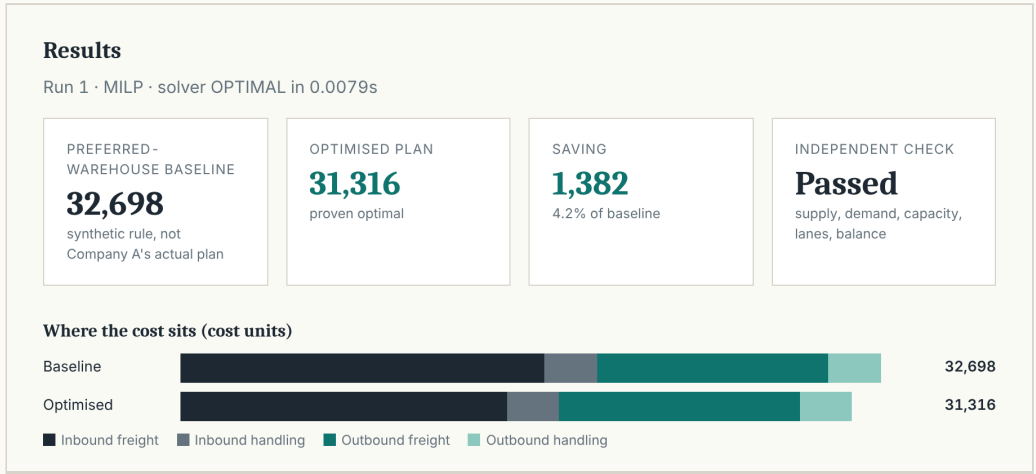
GOAL Lower verified total cost within supply, capacity and lane limits, and keep the final say.

SCENARIO Synthetic demo: 2 suppliers, 3 warehouses, 6 zones, 20 SKUs, 606 pallets a week.

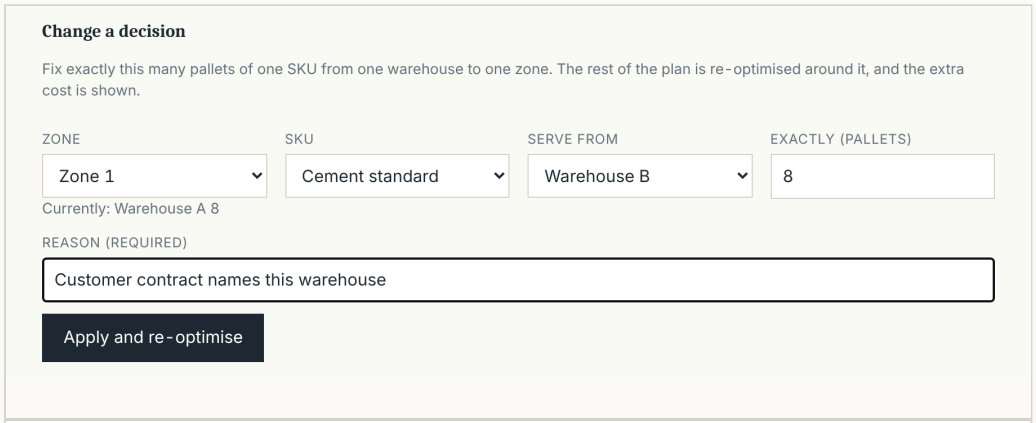
1	2	3	4	5	6	7	8	9
Load data	Data check	Run	Review results	Inspect plan	Read explanation	Change a decision	Approve	Export



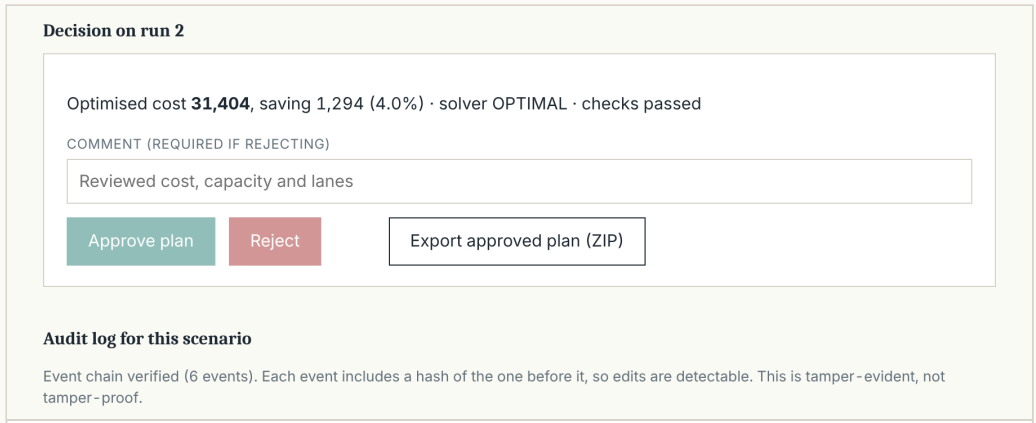
Steps 1 and 2: Data. One click loads the demo, or the planner uploads files. Counts and data checks appear before anything is calculated.



Steps 3 to 5: Results. Baseline 32,698 against optimised 31,316: saving 1,382 (4.2%), solver status and the independent check beside it. 15 decisions differ from the baseline.



Steps 6 and 7: Review and adjust. After approving the explanation, the planner fixes one decision: Zone 1, Cement standard, Warehouse B, exactly 8 pallets, with a reason.



Steps 8 and 9: Approve and audit. The run with that decision (31,404, which is 88 above the free optimum) is approved and locked; export unlocks; each step is logged.

Screens are captured from the running MVP. Figures are synthetic cost units, not a forecast for Company A.

The journey in detail, and what protects the planner

What the planner does, what the system does, and the trust gate at each step. Numbers are from the demo run shown on page 1.

#	PLANNER DOES	SYSTEM DOES	PLANNER SEES	TRUST GATE
1	Clicks "Synthetic demo", or uploads 8 CSV files or one ZIP	Reads the files; ZIPs must be flat, allow-listed names, size-limited, no links	Name, counts (2 / 3 / 6 / 20), 606 pallets demand, 954 capacity	**Safe import**
2	Reads the data checks	Checks columns, numbers, references, supply, capacity and lanes before any solving	"Valid", or errors with file and row; run button hidden until fixed	**Bad data blocked**
3	Chooses MILP (whole pallets) and runs	Builds the baseline, solves for the lowest cost, then verifies both plans with code that does not use the solver	Solver status OPTIMAL in about 0.1 s	**Independent check**
4	Reviews the result	Compares baseline and optimised cost by type; reports warehouse load	32,698 to 31,316; saving 1,382 (4.2%); Warehouse A 100% to 81%, Warehouse C 85% to 100%	**No saving without a valid baseline**
5	Opens the plan and filters changed rows	Lists which warehouse serves each zone and SKU, baseline next to optimised	15 of the zone-and-SKU decisions differ, highlighted	**Nothing hidden**
6	Generates and reads the explanation, edits it, approves it	Computes the facts in code. Default: deterministic summary. Optional: AI wording (Groq), numbers checked	Text tagged "Deterministic summary" or "AI draft", status Draft until approved	**AI never calculates; human approves text**
7	Fixes one decision: Zone 1, Cement standard, Warehouse B, 8 pallets, with a reason	Re-optimises everything else around it and verifies again, as a new run; the first run is untouched	New total 31,404; "88 cost units more than the unconstrained optimum"	**Reason required; cost of overriding is visible**
8	Approves (or rejects with a comment)	Locks the run and writes the audit event; hash-chained log	"This run is approved", event list with timestamps	**Locked after decision**
9	Exports the approved plan	Builds a ZIP: inbound and outbound CSV, comparison, planner decisions, audit log	Download starts; export is refused until approval	**Export only if approved**

Exception: invalid data
A negative rate or missing column is reported with file and row. The run is blocked until the file is corrected.

Exception: unusable baseline
If the baseline rule cannot serve all demand, the saving is shown as "n/a", never as a number. The optimised plan is still valid.

Exception: impossible decision
A planner decision that breaks capacity or lanes is rejected with an explanation, logged, and the current plan stays unchanged.

Exception: no AI or no network
The deterministic summary is used. If AI text contains a number that was not calculated, it is discarded for the deterministic one.

What the journey does not cover: real ERP, WMS or TMS connections (files in and out only), login and roles (one simulated planner), measuring planner time saved or on-time delivery (needs real users).

The what-if tab (close a warehouse, change capacity, scale demand) is an optional side path from step 4.